MCS-221: DATA WAREHOUSING AND DATA MINING

Solved Assignment | July 2026 – January 2027

Course: MCA_NEW / MCAOL **Maximum Marks:** 100

Note: These are model answers for study and assignment preparation. Review them, add course-specific references where required, and make sure you understand the concepts before submission.

Q1. Need, characteristics and choice of data warehouse approach

A data warehouse is a central repository that collects, cleans and organizes historical data from different operational systems so that an organization can perform reporting, analysis and decision-making without placing extra load on day-to-day applications. In the university, admissions, examinations, finance and the learning management system (LMS) may store data in different formats and use different identifiers. A warehouse provides a consistent view of applicants, students, courses, results, fees and online learning activity. It supports trend analysis, accreditation reports, enrolment forecasting, fee monitoring and identification of students who may need academic support. **Characteristics:**

1. **Subject-oriented:** Data is organized around subjects such as students, courses, admissions, finance and academic performance.
2. **Integrated:** Data from heterogeneous systems is standardized using common definitions, formats, codes and identifiers.
3. **Time-variant:** Historical snapshots are retained, enabling comparisons across semesters and years.
4. **Non-volatile:** Warehouse data is generally loaded periodically and queried for analysis rather than frequently overwritten by transactions.
5. **Multiple levels of detail:** It can store detailed records as well as summaries for efficient analysis.
6. **Decision-support oriented:** It is designed for complex queries, dashboards and analytical workloads. **Inmon versus Kimball:** Inmon proposes a top-down approach: build an enterprise-wide, integrated warehouse (traditionally normalized) first, then create departmental data marts. Kimball proposes a bottom-up approach: build dimensional data marts around business processes and integrate them using conformed dimensions and shared facts. For this university, a **Kimball approach** is a practical starting choice if the immediate goal is to deliver useful reports quickly. Separate dimensional marts can be developed for admissions, examinations and finance, using conformed dimensions such as Student, Course, Department and Academic Term. Later, LMS activity can be added while preserving common definitions. This gives users understandable star schemas and good query performance. However, if the university has strong enterprise governance requirements, many complex source systems and a need for a centrally designed integration layer before reporting, Inmon may be preferable. The final choice depends on governance, delivery deadlines, resources and integration complexity.

Q2. Three-tier data warehouse architecture for a healthcare organization

A three-tier architecture separates data preparation, analytical processing and user access. A healthcare organization could use the following flow: **Data sources** → **Staging and ETL** → **Enterprise data warehouse** → **OLAP/semantic layer** → **Front-end tools**

A metadata repository documents definitions and mappings across these components and is used throughout the process. **1. Bottom tier – data acquisition and storage:**

• **Data sources:** Electronic health records (EHR), hospital information systems, laboratory systems, pharmacy systems, billing, insurance claims and medical-device feeds.

• **Staging area:** A temporary controlled area where extracts are landed before loading. It isolates source systems from heavy processing and supports validation and recovery.

• **ETL:** Extracts data from sources, transforms it (standardizes dates, units, codes and patient identifiers; checks quality; resolves duplicates) and loads it into the warehouse.

• **Metadata repository:** Stores source-to-target mappings, table definitions, business terms, lineage, refresh schedules, transformation rules and data-quality information.

• **Data warehouse:** Stores integrated historical data, for example encounters, diagnoses, tests, treatments, costs and

outcomes. Access controls and privacy safeguards are essential for patient data. **2. Middle tier – OLAP server:**
The Online Analytical Processing (OLAP) server organizes data into multidimensional structures or semantic models. Users can analyse measures such as admissions, readmission rates, average length of stay and treatment costs by time, hospital, department, diagnosis or age group. It may use relational OLAP (ROLAP), multidimensional OLAP (MOLAP) or a hybrid (HOLAP) design. **3. Top tier – front-end tools:**
Dashboards, reports, visualization tools, ad-hoc query interfaces and approved analytical applications allow clinicians, administrators and finance managers to explore trends. Role-based access ensures that users see only data appropriate to their responsibilities. **Example:** A hospital manager can compare monthly readmission rates across departments, drill down to a particular hospital unit and examine whether trends vary by diagnosis. The architecture provides consistent, timely information while keeping analytical workloads separate from operational care systems.

Q3. Dimensional model for supermarket sales

A **star schema** is suitable because supermarket managers need fast, understandable analysis of sales by month, store, product and customer. The central fact table records measurable business events, while dimension tables describe the context of each event. **Fact table: FactSales**
Suggested grain: one row per product line on a sales transaction (one product sold at one store at a particular time). Fields include:
• DateKey (foreign key)
• ProductKey (foreign key)
• StoreKey (foreign key)
• CustomerKey (foreign key, possibly an anonymous/unknown customer key)
• PromotionKey (optional foreign key)
• TransactionID (degenerate dimension or transaction identifier)
• QuantitySold, UnitPrice, DiscountAmount, SalesAmount, CostAmount and ProfitAmount. SalesAmount may be calculated as quantity × unit price minus discount. ProfitAmount can be calculated from sales revenue minus cost, provided the cost definition is consistent. **Dimensions:**
• **DimDate:** date, day, week, month, quarter, year, holiday indicator.
• **DimProduct:** SKU, product name, brand, category, subcategory, supplier.
• **DimStore:** store, city, region, store type and floor area.
• **DimCustomer:** customer segment, age band or loyalty status, subject to privacy rules. Avoid storing unnecessary personal details.
• **DimPromotion:** promotion type, campaign and discount category. **Schema and justification:** The FactSales table is in the centre, linked to each dimension through keys, forming a star. This reduces the number of joins and makes business queries easier. A snowflake schema could normalize product category or geographic hierarchies, but the extra joins may make queries less intuitive. For the stated needs, a star schema is a strong default. **Analysis examples:**
• Monthly sales: sum SalesAmount grouped by year and month.
• Customer behaviour: frequency of purchases, average basket value and product-category preference by customer segment.
• Product performance: quantity, revenue and profit by SKU, category and brand.
• Store comparison: compare revenue and margin across regions. For reliable analysis, the grain must be fixed, measures defined consistently, returns handled (for example with a separate FactReturns or negative sales rows), and slowly changing attributes such as customer segment managed with an appropriate history policy.

Q4. ETL process and data quality issues

ETL means Extract, Transform and Load. It brings data from e-commerce, ERP and CRM systems into a consistent warehouse model. **1. Extract:** Identify required tables, fields, refresh frequency and extraction method. Data may be extracted through database queries, APIs, change-data capture or scheduled files. Initial full loads may be followed by incremental loads based on timestamps or change logs. Extraction should minimize impact on operational systems and preserve source identifiers. **2. Stage and profile:** Land raw extracts in a staging area with batch IDs and timestamps. Profile values to detect missing fields, invalid ranges, duplicate records, unexpected codes and distribution changes. Keep enough audit information to trace errors back to the source. **3. Clean and transform:**
• **Missing values:** recover from a trusted source where possible; otherwise use an explicit “unknown” category, a

suitable imputation method or exclude the record when justified. Do not silently treat missing values as zero.
• **Duplicates:** use business keys and matching rules to identify duplicate customers, orders or products; merge only when evidence supports the match.
• **Inconsistent formats:** standardize dates, time zones, currency, address formats, units and case.
• **Conflicting codes:** map source-specific product, region and status codes to common warehouse definitions.
• **Entity resolution:** match the same customer across CRM and e-commerce using reliable identifiers and controlled fuzzy matching where appropriate.
• **Invalid values:** apply domain/range checks and refer exceptions for correction rather than inventing values.
• **Derived fields:** calculate net sales, tax, margin and order duration using documented business rules.
• **History:** use slowly changing dimensions where historical changes in customer, product or organization attributes need to be retained. **4. Integrate and validate:** Join related entities, enforce referential integrity, reconcile totals against source systems and run quality checks such as row counts, uniqueness, valid key checks and financial balance checks. **5. Load:** Load dimensions first when needed, assign surrogate keys, then load fact tables. Use transactions, idempotent batches and restartable jobs to avoid partial or duplicate loads. Refresh aggregates and indexes where appropriate. **6. Monitor:** Record errors, rejected rows, lineage, load duration and quality metrics. Alert owners when thresholds are exceeded and retain rejected records for remediation. A reliable ETL pipeline is repeatable, documented, secure and auditable. Data quality rules should be agreed with business owners because transformations can otherwise change the meaning of the data.

Q5. OLTP, OLAP and OLAP operations

OLTP (Online Transaction Processing) systems run day-to-day transactions. They handle many short inserts, updates and lookups with strong consistency and current data. Examples include supermarket checkout, bank transfers, hospital registration and online order placement. Their databases are commonly normalized to reduce duplication and support reliable transactions. **OLAP (Online Analytical Processing)** systems support complex analysis across large amounts of historical data. They perform aggregations and comparisons for planning and management. Examples include analysing sales by month and region or comparing hospital costs across years. OLAP systems often use warehouse/star-schema models, summaries and read-optimized structures. **Main differences:**
• Purpose: OLTP operates the business; OLAP analyses business performance.
• Data: OLTP is mainly current and transaction-level; OLAP is integrated and historical.
• Queries: OLTP uses short, predictable queries; OLAP uses complex joins, grouping and aggregation.
• Workload: OLTP has frequent concurrent writes; OLAP is mainly read-heavy.
• Design: OLTP commonly uses normalized models; OLAP commonly uses dimensional models.
• Users: OLTP is used by operational staff and applications; OLAP by analysts, managers and executives. **OLAP operations:**
1. **Roll-up:** Summarize to a higher level, such as daily sales → monthly sales → yearly sales.
2. **Drill-down:** Move from summary to detail, such as annual sales → quarter → month → day.
3. **Slice:** Fix one dimension value, for example view sales for 2026 only.
4. **Dice:** Select a sub-cube using several conditions, such as sales for two regions, selected product categories and Q1–Q2.
5. **Pivot (rotate):** Reorient the view, such as switching rows from regions to product categories and columns from months to years. These operations let managers identify trends, compare performance, investigate exceptions and evaluate actions. For example, a roll-up shows that quarterly revenue has declined; drill-down locates the affected month; slice and dice isolate a region and category; pivot compares stores and product groups. OLAP informs decisions but does not, by itself, prove causation.

Q6. Data preprocessing for customer data

Data preprocessing improves the quality and suitability of data before mining. A financial institution must handle customer data carefully because errors can lead to unfair credit decisions, regulatory issues and privacy risks. **1. Data cleaning:**
• Missing values: check whether the value is genuinely unavailable or not applicable. Impute numeric values using a suitable method (such as median for skewed data) or use a missingness indicator when informative. Use a separate category for missing categorical values when appropriate.

• Noise and outliers: detect unusual values with range checks, box plots, robust statistics or domain rules. Correct confirmed errors; retain genuine extreme cases where they represent real customers, and consider robust scaling or models resistant to outliers.
• Inconsistency: standardize date formats, currency units, customer status codes and spelling. Validate age, income, loan balance and payment dates against business rules.
• Duplicates: identify repeated customer or account records and reconcile them using reliable keys. **2. Data integration:** Combine data from account systems, loan applications, payment histories and CRM. Resolve identities, align definitions, remove redundant attributes, standardize keys and document source lineage. Check that the same metric means the same thing across systems. **3. Data transformation:**
• Normalize or standardize numeric variables when required by the algorithm (especially distance-based methods and SVM).
• Aggregate transaction records into meaningful features, such as missed payments in the last 12 months or debt-to-income ratio.
• Encode categorical variables using one-hot or other appropriate encoding.
• Apply log or other transformations to highly skewed variables when justified.
• Use feature construction with clear definitions and prevent data leakage by using only information available at the prediction time. **4. Data reduction:** Reduce size while retaining useful information through feature selection, dimensionality reduction (such as PCA), sampling for exploration, aggregation or removal of redundant attributes. For regulated credit models, preserve interpretability and validate that reduction does not harm important subgroups. **5. Discretization:** Convert continuous values into intervals, such as age bands or income ranges. Equal-width, equal-frequency and domain-informed binning are possible. Bins can improve interpretability but may lose information; cut points should be selected using training data only. Finally, split data into training, validation and test sets using a time-aware strategy if predicting future default. Fit imputation, scaling, encoding and discretization on training data only, then apply them to validation/test data. Assess accuracy, calibration, fairness, stability, privacy and explainability before deployment.

Q7. Association rule mining and Apriori

Association rule mining discovers items that frequently occur together in transactions. A rule has the form $X \rightarrow Y$, where X and Y are itemsets and $X \cap Y = \varnothing$. Consider five supermarket transactions:
T1: {Bread, Milk}
T2: {Bread, Diapers, Beer, Eggs}
T3: {Milk, Diapers, Beer, Cola}
T4: {Bread, Milk, Diapers, Beer}
T5: {Bread, Milk, Diapers, Cola} Suppose minimum support is 40% (at least 2 of 5 transactions) and minimum confidence is 60%. **Measures:**
• **Support($X \rightarrow Y$)** = transactions containing $X \cup Y$ / total transactions.
• **Confidence($X \rightarrow Y$)** = support count($X \cup Y$) / support count(X). It estimates how often Y appears when X appears.
• **Lift($X \rightarrow Y$)** = confidence($X \rightarrow Y$) / support(Y). Lift above 1 indicates positive association; 1 indicates independence; below 1 indicates a negative association under this measure. For the rule **Diapers** $\rightarrow$ **Beer**, Diapers appears in T2, T3, T4 and T5 (4 transactions); Beer appears in T2, T3 and T4 (3); both appear in T2, T3 and T4 (3). Therefore support = 3/5 = 60%; confidence = 3/4 = 75%; lift = 0.75 / 0.60 = 1.25. In this small illustrative dataset, diaper purchases are associated with beer purchases more often than the baseline beer rate. This is not proof that one causes the other. **Apriori algorithm:**
1. Scan transactions to count individual items and keep those meeting minimum support (L1).
2. Generate candidate 2-itemsets from frequent 1-itemsets and count them (C2); retain frequent sets as L2.
3. Join frequent (k–1)-itemsets to create candidate k-itemsets.
4. Prune any candidate if one of its (k–1)-subsets is infrequent. This uses the Apriori property: every subset of a frequent itemset must also be frequent.
5. Scan data to count remaining candidates and repeat until no new frequent itemsets remain.
6. Generate rules from frequent itemsets and retain rules that satisfy confidence or other chosen thresholds. Apriori can be computationally expensive when there are many items or candidates. Alternatives such as FP-Growth avoid explicit candidate generation in many cases. In practice, analysts should consider transaction size, business relevance, minimum thresholds and whether associations are stable over time.

Q8. Classifier comparison and credit-risk recommendation

Classification predicts a categorical outcome, such as whether a borrower will default. The model should be evaluated on unseen data and selected using business, regulatory and fairness requirements—not accuracy alone. **1. Decision tree:** Splits data using feature-based rules. It is relatively easy to visualize and explain, handles nonlinear relationships and needs limited scaling. A single tree can overfit or be unstable; pruning and depth limits help. Ensemble trees may improve performance but can be less transparent. **2. Naïve Bayes:** Uses Bayes' theorem and assumes features are conditionally independent given the class. It is fast, works well with high-dimensional data and produces probabilistic predictions. Correlated financial features can violate the assumption, and probability estimates may need calibration. **3. k-Nearest Neighbour (k-NN):** Predicts from the classes of nearby training examples. It is simple and can model complex local patterns, but requires feature scaling, can be slow at prediction time, is sensitive to distance choices and noise, and struggles with many dimensions or imbalanced data. **4. Support Vector Machine (SVM):** Finds a separating boundary with a maximum margin; kernels can model nonlinear boundaries. SVMs can perform well with high-dimensional data, but scaling and parameter tuning are important. Large datasets may be costly, and explaining a complex kernel model is difficult. **Recommendation:** For credit-risk prediction, a **pruned decision tree** is a defensible initial recommendation when interpretability and auditability are priorities. Its rules can be inspected by risk officers and explained to stakeholders. However, it should not be assumed to be the most accurate in every dataset. A practical project should compare it with regularized logistic regression and the listed alternatives using time-separated validation, ROC-AUC or PR-AUC, recall at an acceptable false-positive rate, calibration, stability and subgroup fairness metrics. If predictive performance is the overriding goal, an SVM or ensemble may outperform a single tree, but additional explanation and governance may be needed. The model must avoid leakage, use only variables available at the decision point, protect sensitive data, document feature definitions and thresholds, and be monitored for drift. Credit decisions should include appropriate human review and compliance with applicable lending and anti-discrimination requirements.

Q9. Clustering techniques, K-means and DBSCAN

Clustering groups similar records without using predefined class labels. It can be used for customer segmentation, geographic analysis, anomaly exploration and grouping products. **Partitioning methods:** Divide data into a specified number of clusters, with each record assigned to a cluster. K-means is a common example. It is efficient for large numeric datasets but requires choosing K, is sensitive to scaling and outliers, and works best for roughly spherical, similarly sized clusters. **Hierarchical methods:** Build a hierarchy of clusters. Agglomerative clustering starts with individual points and repeatedly merges clusters; divisive methods split larger clusters. The result can be shown as a dendrogram, and the number of clusters can be chosen later by cutting the tree. It is useful for exploring nested structure but can be computationally expensive and early merge/split decisions may be difficult to undo. **Density-based methods:** Identify dense regions separated by sparse regions. DBSCAN is a well-known method. It can find arbitrarily shaped clusters and label isolated points as noise, but it is sensitive to the choice of parameters and can struggle when cluster densities vary greatly or dimensions are high. **K-means working:**

1. Choose K initial centroids.
2. Assign each point to its nearest centroid, usually by Euclidean distance.
3. Recalculate each centroid as the mean of the points assigned to it.
4. Repeat assignment and update until assignments or centroids stop changing significantly, or a limit is reached.

K-means minimizes within-cluster squared distance and can converge to a local optimum, so multiple initializations are advisable. Scale numeric variables and investigate outliers before use. **DBSCAN working:** Select ε (neighbourhood radius) and MinPts (minimum number of nearby points). A point with at least MinPts points in its ε-neighbourhood is a core point. Core points connected through density reachability form a cluster; border points are attached to a cluster but do not have enough neighbours to be core points themselves; points not assigned to any cluster are noise. **Comparison for noisy data:** K-means is fast and useful when clusters are compact and K is known, but outliers can pull centroids and every point is assigned to a cluster. DBSCAN is often better when noise and irregularly shaped groups are expected because it can leave outliers unclustered. However, it may miss clusters with very different densities and needs distance/parameter tuning. For customer segmentation with clean, compact groups, K-means may be suitable; for spatial hotspot detection with noise, DBSCAN may be preferable. Evaluate cluster quality and domain usefulness rather than relying on one score alone.

Q10. Emerging trends in data warehousing and data mining

Modern data warehousing and mining combine scalable storage, distributed processing and advanced analytical techniques to provide faster, more useful business intelligence. **1. Cloud data warehouses:** Cloud platforms offer managed storage and compute that can scale independently, support elastic workloads and reduce the need to manage physical infrastructure. They enable collaboration and integration with BI and machine-learning services. Organizations must still manage costs, access controls, data residency, encryption, vendor dependence and workload optimization. **2. Big data analytics:** Distributed storage and processing frameworks handle high-volume, high-velocity and high-variety data such as transactions, logs, clickstreams and sensor feeds. Parallel processing enables large-scale aggregation, pattern discovery and predictive analysis. Data governance and quality are critical because large volumes do not automatically mean reliable insight. **3. Text mining:** Applies natural-language processing and statistical or machine-learning methods to unstructured text, including customer reviews, support tickets, survey responses and documents. Tasks include sentiment analysis, topic discovery, classification, entity extraction and summarization. Businesses can identify recurring complaints, product requests and emerging risks. Privacy, language variation, bias and context must be considered. **4. Web mining:** Extracts useful patterns from web content, link structures and usage logs. Content mining analyses page text and media; structure mining studies links; usage mining examines clickstreams and navigation. It can support recommendations, website optimization, customer journey analysis and personalization. Consent, privacy, bot traffic and changing user behaviour affect reliability. **5. Data stream mining:** Analyses continuously arriving data, such as payment events, IoT sensors, website events and fraud signals. Instead of repeatedly processing the entire dataset, stream systems maintain summaries or update models incrementally and may use windows to focus on recent data. This enables near-real-time alerts, anomaly detection and operational response. Challenges include high throughput, delayed events, concept drift and false alarms. **How these enhance business intelligence:** Cloud platforms provide flexible infrastructure; big-data tools make large and varied datasets practical to analyse; text and web mining reveal insights from unstructured and behavioural data; stream mining supports timely action. Together they enable integrated dashboards, more detailed customer understanding, demand forecasting, fraud detection, process optimization and evidence-based planning. Successful adoption requires clear business questions, data governance, security, quality monitoring, responsible AI practices and measurable outcomes. Technology should be chosen to solve a defined problem rather than adopted only because it is new.

End of model answers. The assignment notice states that all ten questions are compulsory and that the remaining 20 marks are for viva voce. Check the original assignment instructions and your study-centre/LMS submission requirements before submitting.